Your Inner Cosmos with David Eagleman

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So Gary did it again, got us another neuroscientist. But this one has got some unorthodox means, pushing people off of rooftops. Yeah. Just to see how their brain works. What's up with that? How to make memories, Neil? I think the memory is me crapping my pants. Coming up on Star Talk, your inner cosmos. Welcome to Star Talk, your place in the universe where science and pop culture collide. StarTalk begins right now. This is StarTalk Special Edition. I'm Mule the Grastitian, your personal astrophysicist. And if it's Special Edition, it means Gary O'Reilly is sitting right there. How you doing? Exactly! With those cheers or you exhaling into the mic a little a bit of both I am I'm a smoker. So yeah, yeah, that is Chuck nice cheering on Gary O'Reilly. Yes. So Gary, we've got more neuroscience back in one of our favorite subjects. And we can't get enough of it. I think the field is, you know, relative to other branches of science, it's relatively new. And so Every few months we get a new neuroscientist on telling us about new discoveries, new understandings, and I love every minute of it. So what have you stitched together for today? Ah, right. Let me explain it to you this way before we reveal who our guest might be. Often on Star Talk we explore the outer cosmos, and today we'll explore our inner cosmos. The human mind is flexible. What are the boundaries though of what our minds can do? Imagine if you could see with your tongue or hear through your skin. The reality is, we can. And we'll get to that later on. Now, what is neuroplasticity? I guess if I can even say it. Now ask yourself, why do we dream and follow it with do we perceive time correctly? Now that is a question, but that's enough for now. So Neil, let's bring on our guest and we'll get straight to it. Well, and our guest in 2020, published a book live wired, the inside story of the ever-changing brain. Or is that really our ever-changing understanding of our non-changing brain? We'll find out right straight away on that one. Please join me in welcoming David Eagleman. Did I pronounce your name right correctly there? Yes, you did. David Eagleman, excellent. Your professor?

at Stanford University specializing in neuroscience. And I love it. We got people rising up trying to have us understand ourselves. But we got to start out, because I heard Gary say, we're seeing with our tongues. Are we snakes

here? Let's just start there, and we'll pick it up from wherever that leaves off. OK. Well, great. Great to be here and see you guys again. the issue is your brain is locked in silence and darkness in the vault of the skull and it doesn't have direct access to anything in the outside world so it's not seeing photons or hearing sounds it's just converting photons into electrical spikes that plunge Into the darkness of your brain and you convert air compression waves into spikes that plunge into a different part of the brain and so on and all you have are all these signals running around and the brain from that constructs your whole technicolor subjective.

world. So an idea that I got interested in a couple decades ago in my lab is could we push information into the brain via different channels? And will the brain figure it out? Because it's all made of the same currency. It's all the same stuff. And so it turns out this has a long history. The first paper on this was In 1969, in the journal Nature, a scientist named Paul Bocchi Rita put blind people into a modified dental chair, which had a little grid of solenoids that poked people in the back. So imagine, let's say, 8 by 8 little grid where you feel things poking you in the back. And he set up a video camera, and whatever the camera was looking at, you would feel that in your back.

So it's like these games that you do as a kid where you draw something with your finger on someone's skin and they try to figure out what it is. It's the same idea. And he was able to show that blind people get really good at being able to perceive what's in the world out there just based on the skin of their back. And it turns out that the history of this even I discovered more recently that it goes back to the 1880s where people were first trying this sort of thing about turning light into patterns of vibration on the forehead or something. So now what we have is the capacity to really miniaturize these electronics and make this work. And so one of the things I've worked on in my lab for years is converting sound into patterns of vibration on the skin so that deaf people can hear that way. It's the same information. It's just getting up to the brain via an unusual route through the skin. By the way, in the film, in the film Immortal Beloved, which was basically chronicled Beethoven in his most productive years, including his transition to deafness. There's a beautiful scene, which I didn't fully understand when I first saw it. He's there alone with his piano in this big open room, and his head is resting

on the top of the piano while he's banging out these random sounds on the keys. And we were not explicitly told at this point, that he was deaf. I think if you knew he was, you would know this, but I didn't, first time seeing the film. And just these very cacophonous notes being banged out. Again, his head is resting on the top of the piano. And after 30 seconds of this outcomes, these beautiful music, and then you, oh my gosh, he can't hear it. He's feeling the vibrations into his cranium.

through the cavity of the piano itself. And for me, to this day, it's one of the most poignant cinematic moments ever. And that's him, I'm just repeating what you're saying, it's him converting the vibrations, external vibrations into his skin and getting into his cranium. There's an old history to this actually. So people who are deaf and blind use this method called Tadoma where they put their hand on the speaker's face as in the person speaking to them, put their hand on their face, and their hand is covering different parts of them so they can feel what's going on, the vibrations in the throat and the muscles of the jaw and so on. And by pulling all these different pieces of data on their fingertips, they can figure out exactly what the person is saying. Let me ask you, what you just talked about with the skin, makes me think about when blind people read Braille. That signal, is it still activating in the occipital lobe and then the occipital lobe actually takes over or takes part in the sensory of a haptic experience or is it something else where different parts of the brain are being activated to compensate?

What exactly is happening? Okay, lovely question. So let's look at this. So when you are reading a book and your eyeballs are landing on the page at different moments, you're seeing these squiggles on the page that your brain makes sense of and you are so overtrained on this that you're not even aware of these squiggles unless you look, for example, at a foreign alphabet. In which case when you look at the foreign alphabet you can't believe how weird these things are and that anybody can ever see this. Yeah, it's all squiggles. Yeah, but when it's your own language you say my god that's the most obvious thing and and all that emerges is the the meaning of the text and you laugh and you cry at the trials of the the protagonist okay.

Now, when a blind person is reading Braille, they're passing their fingers over these bumps. As you said, Neil, they're just each letter, but they're not thinking of it as letters. As they pass their finger over, they're laughing and crying because the meaning emerges. And it's a great example. I'm glad you brought this up about Braille because it shows how arbitrary it is. You know, we're using these giant spheres embedded in our skull to pick up on photons to get meaning that way, but a blind person gets it through the fingertips that way. So let me ask you this. If the brain doesn't care, I'll say it like that, which way it receives the data, what does it tell us about our brain's ability to adapt to this circumstance and our perception in general?

Okay, so actually actually let me back up just to one more issue that Chuck asked which is this question of When you're feeling braille with your fingertip, is that going to the occipital cortex at the back of the brain, which is normally where we see? The answer is sort of. What happens when you're using your fingertip is it goes to the somatosensory cortex because that's where your fingertip directly goes. But the key is that the brain doesn't let any land lie fallow. And so if your visual cortex is not getting used by, you know, there's hundreds of things that could go wrong with your eyes and your optic nerves or whatever. If that is no longer getting used, then that territory gets taken over by the Braille reading. And you can demonstrate this because, well actually many ways, even in the past, you can demonstrate it that if a blind person uses Braille and gets a stroke in their occipital cortex at the back of their head, they get worse at reading. And nowadays we do it using fMRI, brain imaging, where we look at what's going on when you're using Braille.

you see that the visual cortex, what we would have normally thought of as the visual cortex getting activated. But that's because it's because the brain doesn't let any territory go to waste. So if you go deaf, the parts of your brain that would have taken care of hearing get taken over. And that's why deaf people are really good at seeing other things. For example, they can see your accent on your lips as you're moving and talking. And of course, blind people can do all kinds of extraordinary things because all that territory, the visual cortex has been taken over. So they can hear exactly things about your footfall, or they can localize things that hearing people can't do so easily. So much so, I actually saw a piece years ago about a

young kid who was blind and developed echolocation as a means of moving around by using clicks in his mouth and being able to, what we would call, see the room, but of course not in the sense that we consider sight. Right. The young man you're referring to is a kid named Ben Underwood who unfortunately passed away a few years later.

But I actually had his mother come in to my class at Stanford that I teach on brain plasticity to talk with us about this. What's fascinating, you're exactly right. So Ben makes these clicking sounds and he's getting the information back. I saw a documentary on him that said something like, Ben is the only person in the world that can use echolocation. But it turns out that's not true. tens of thousands of blind people use echolocation and in fact the term was coined in a 1930 science paper in the journal science that was called echolocation in bats and the blind. So people use is they use either tapping their stick or these clicks with their mouth or they, you know, make a sound of some sort or a snapping sound with their fingers and they get the echoes back and they're able to construct. It's very low resolution, of course, but they're able to construct something about what's going out there in front of them. That's why. This is Ken, the nerd neck Zabara from Michigan, and I support Star Talk on Patreon. This is Star Talk Radio with Neil deGrasse Tyson. Why doesn't?

this land grab of the spheres take place in someone with a full suite of senses? The answer is all your senses are in competition with each other all the time and the reason all your senses stay intact is because they're all fighting for it and they're getting data in there. Now it depends a little bit on what is relevant to you as a person. If you go into Let's say perfumery, then your olfactory cortex dealing with smell is going to enhance and get bigger. If you are a microscopist who needs to stare at things all the time, that might be bigger. If you're a musician who needs to really hear very subtle details of the music and the orchestra, then that's going to dominate a bit. But in general, you're getting all of these senses all the time. Wow. Now, this led me to a whole new theory about why we dream.

You guys want to hear that? Oh, okay. Yeah. Oh, yeah. Okay. So it turns out that because there's this competition with all the senses going on, one of the new pieces of data just let's say 14 years ago that we got was that the

takeovers can happen pretty rapidly between the senses. So for example, if you take people My colleagues at Harvard did an experiment where they blindfold people tightly and they stick them in the scanner and they're measuring things as they make a sound or they touch them on the hand, things like that. It turns out that after 60 minutes you start to see activity in the occipital lobe, what we normally think of as the visual cortex based on touch or sound.

We're seeing activity in the visual cortex just because the person has been blinded normally if you touch someone on the hand you see that activity in a particular part called the cemetery cortex if you play a sound you see activity in the auditory cortex but now you start to see a little bit of activity showing up in the visual cortex. if you've blinded them for 60 minutes. And so I was really struck by how rapidly the takeover starts. And so that's what led me and my student, Don Vaughn, to a new hypothesis. We're on a planet that rotates into darkness for half the time. And when you're in the dark, you can still smell and touch and taste and hear, but you can't see. And obviously I'm talking about evolutionary time, not the last nanosecond of electrified existence that we have. And so That puts the visual system at a real disadvantage, and when you sleep, if there's a bug crawling on your skin or a smell, that'll wake you up. You're still sensing that, but you can't see anything.

The visual system is at a real disadvantage. And so what we realized is dreams are the brain's way of keeping that territory active every 90 minutes. You have this random burst of activity that just goes blasting into the primary visual cortex. And that's it. When you dream, you've got this very ancient circuitry that does nothing but blast random activity into the primary visual cortex. So that's amazing. So basically the brain creates its own screensaver to occupy the real estate from invasion. Here we go. We've gone all game of thrones. So we've got invasion. That's actually cool, man. Yeah. So the courtesies are always in competition. And when we go to sleep, it is the competition that fears that this area of the brain says, hey, you know, it's like when you work in radio, you can't go on vacation.

Because when you come back, you don't have a job. Yeah. No, seriously. That's the rule in radio is like, don't go on vacation. Because when you

come back, somebody would be behind that microphone that started his career on radio. Yes. Okay. So the answer is yes, the competition is very fierce. But because we live on this planet that has this rotation, only one sense is disadvantaged. Now, if we were a tidally locked planet or something where we were always facing our sun, we would, I think, not have dreaming. There would be big differences in the way that our brains wired up. But because we're in the situation where you have to protect it, from takeover every single night, yep, you can't go on vacation. Well, it's not just whether the sun is set, it is whether your eyes are closed, right? I mean, when you sleep, whether or not it's day or night, doesn't that matter too? So it is the case that currently people have night shift jobs and whatever, and so things can be different. But what we're talking about in the development of the brain is over many hundreds of millions of years and so it was when the planet rotates into darkness that's when almost all animals sleep and Part one of the hypotheses for why we sleep is just to stay out of trouble It's actually easier just to curl up in the corner of a cave and and shut down and do all these other things like energy Restoration and so on but it's both it's both that it's pitch black outside on most nights and it's you know and you sleep and so uh you need a way of defending that territory yeah and wouldn't some of that be connected to your circadian rhythm because for instance i wake up normally at first light and i didn't realize why for the longest time i still don't know why but then i bought a blackout mask And so when I wake up, the first thing I do, I'm wake, right? Like my body's like it's time to get up. And I put on the blackout mask. I go back to sleep with no problem. And as a matter of fact, that's when I have to set an alarm because I will sleep much longer than I would like because it's just black. It's just darkness. But so it's like almost like even though my eyes are closed, My body is still able to sense the light that is coming into the room at first light. Or maybe this is all just me. I'm really asking you a question here.

Yeah, no, that's right. We we've traditionally been totally tied to what's going on with with light and darkness In a way that we've you know drifted from now because we have artificial light anywhere We want that sort of thing but yes what what I'm talking about in the brain are these very very ancient Systems and in fact this circuitry that causes these are called PGO waves Pontine geniculate occipital doesn't matter the details, but these

Essentially, you find this across all animal species, all animals dream. As far as we can tell, as far as we can measure dream-like activity and so on, all animals are doing this because it's so fundamental and ancient. It's sort of, as far as I can tell when I'm looking at the anatomy, it's like an add-on circuit of, oh, shoot, we've got this extra problem because of the rotation of the planet. Let's just go ahead and blast activity in there to save this.

So one of the things we did is we measured very carefully across 25 species of primates, which have different amounts of brain plasticity, how flexible their brains are. So humans are unbelievably We essentially drop in the world half baked and we absorb everything around us, our language, our culture, all the science that's come before us. We absorb all that and then we springboard off that to the next level. But even among primates, there are those that come out that are not particularly flexible. They're sort of pre-programmed. You can think about it that way. And so anyway, we looked across all these primate species with the hypothesis that the more flexible the brain, the more brain plasticity, the more dreaming will happen every night. And that's exactly what we find is a significant correlation where if your brain has more danger of big shifts happening, more danger of takeover of the occipital cortex, then you have more dreaming. But if your hypothesis is accurate as a general statement about human physiology and our sensory system, it seems to me you could do this with other senses as well. So if you plug your ears, denying any auditory signals from entering your brain. That's a part of your brain that's now not being used. Would you have more auditory dreams as opposed to visual ones under those conditions?

So this is exactly what happens when people go into, let's say, solitary confinement. For example, at Alcatraz, I interviewed some people who had been in the hole locked up in Alcatraz. This is exactly what happens. You have auditory and visual hallucinations because your brain is fighting back and saying, wait a minute, I'm used to getting these signals and now I'm not getting any of this information. So it starts generating it on its own. I've actually, I've also hypothesized that Tinnitus which is ringing in the ears is another expression of this what I call the defensive activation theory Which is your brain is used to getting all these frequencies that it's hearing but when people's The cells in their inner ear die off in some range their brain

says wait a minute I'm not hearing that frequency anymore So it starts generating it on its own starts making its own version of it and that's what the ringing in the ears is damn That sucks. Yeah. So going back to dreams, David, this is a two-part question. What, A, if we do not dream, what is happening there? And B, why is it we don't always remember our dreams? So great question. Everybody dreams, and that leads to part B, which is that Often you don't remember your dreams and that has a lot to do with when you wake up during the night So what happens is you have different stages of sleep during the night and dreaming happens at the lightest stage of sleep and Often people will wake up right at that stage and they'll remember it just fine And of course, there are ways to remember it even better, which is you write it down or you speak it into your cell phone and then you can recall it. But the key is when you are dreaming, the brain is smart enough to say, hey, I don't really need this. It's random stuff, which was why it's not interesting to hear about other people's dreams because it's just random garbage. I mean, I call it sticking my head in the night blender. I hate dreaming. It doesn't interest me at all the content of it, but.

Let me say two things. The brain is a storyteller. So if you blast random activity in there, it's going to make stories out of it, make narratives. Usually those narratives follow the connections that are hot from the day. So things that happen in your day will show up in your dreams, but there's this very loose association and bizarreness and typically magnified emotion too. Anyway, All these things happen in a dream, but your brain is smart enough to know that it's not really that useful to remember that. So your memory systems are mostly shut down during dreaming. And that's why when you wake up, you might remember a dream for, let's say, 10 minutes, 15 minutes. You say, oh my gosh, this happened. My heart's pounding, all this stuff. But then it kind of goes away. And here's the really interesting thing. That gives you insight into what it's like to have, let's say, Alzheimer's disease.

because somebody with Alzheimer's, they have short-term memory. They remember what just happened, you know, a few minutes ago. But after call it 15 minutes, it just fades away. So they'll get off the phone. They'll say, OK, I'll meet you there in a few minutes. And they hang up the phone. And then it just fades away the same way that a dream fades away for you. It's

like Gossamer that just disappears. David, you mentioned consciousness. And as you're explaining these things, it sounds to my mind, which is obviously limited, that the brain will only manufacture and cement memories during its conscious state. Because while we're dreaming, we're not quite conscious, so it doesn't quite imprint and hold on to them. Is that kind of right?

That's generally the case. And by the way, your memory is terrible. I mean, most things that happen to you, you don't remember. You don't remember. Yeah. Yeah. Because memory, it's like a sieve. And that's because most stuff passing through your system just isn't relevant to you for whatever your goals are. You don't remember the number of cracks in the sidewalk or the shirts that people are wearing or the number of trees or what. It doesn't, none of that matters, even though that hit your sensory systems. Yeah, most of the sensory information we take in is garbage. Is that not our brain editing? Is that our brain saying, need it, don't need it, don't need it? No, it's not. It's an interesting question because it's not that your brain takes in all this stuff and figures out what to get rid of. It's that it's not taking in most everything. For example, you've probably seen these things like the invisible gorilla experiment. Have you seen that? Yeah, it's brilliant.

It's completely brilliant. But there's a million examples like this. I only mentioned that one because maybe that's what we all know in common. But the point is you're watching these people pass the basketball. The guy in the gorilla costume comes in and you just don't see it. It's not that your brain says, oh, there's a guy in a gorilla, but that's not relevant, blah, blah. It's that most of the stuff hitting your retinas, you don't see it. You don't see it. Just to catch people up on this. So the gorilla experiment is a video and you are tasked as a viewer of the video to count how many times a basketball Changes hands in that video. So you're focusing on this man. They're like three or four or five people passing the ball around Meanwhile a person in a gorilla suit enters the scene looks around and then exits the scene and then at the end they ask you did you see the gorilla and It's like no there was no gorilla there and then you they were showed again and there it was the gorilla waves at you and You just weren't paying attention Is that a result of the fact that we are focused on something that our brain is trained on? Or is

it because we are literally building what we want to see in our brain as we look at any scene?

Yes, both that's it's an excellent question, but here's the thing the gorilla is hitting your retina I mean all all those photons are there and if you were asking a different question like is there a gorilla in the scene they'd have no problem seeing it there even if your eyes were doing the identical motions and so on It's because all we ever do is match our internal model, all the stuff happening inside the brain, to the little bits of sensory data that come dribbling in through your eyes, your ears, your fingertips, so on. Your brain is building a model of the world and all it's trying to do is say, hey, is that accurate? And by the way, Under different circumstances will rely more on your internal model or or more on the outside world So if you are in your own home, you're essentially operating entirely with your internal model about where things are in the furniture and so on but if you go to a new Country and you're walking around the streets. You're having to pay attention more to the outside But it's pretty rarely that we do that mostly we fall into routine of various sorts and therein is a profound source of bias Right testimony. Yes Hold on, did you say eyewitness testimony as I said? Yeah. Okay, so that involves much more too because that involves the fungibility of memory. You believe that you saw something and obviously what happens with eyewitness testimony is something important and interesting is happening over there. So you look at what's happening, but it's unbelievable. Memory is not like a recorder where you're just pulling something back out.

Instead it's a reconstruction and it's influenced by lots of things including what other people tell you and so in fact I'll just mention really quickly the some colleagues of mine did a great study on this right after 9-11, 2001, they found a bunch of New Yorkers who had seen the towers collapse and they quizzed them on what exactly they saw on September 11th, and then they were clever enough to also ask what they saw on September 10th, what they did, what they ate for lunch on September 10th, and so on. Then they found these people again a month later, three months later, a year, and they did finally a follow-up 10 years later. And it turns out that Both memories both the banal memories from September 10th and the extremely emotionally intense memories from September 11th They all drift they drift

around Entirely and the only thing that stays constant is the stuff from September 11th that you know that you saw in the news like okay There were two planes they hit the towers at this time that stayed constant because there was a constant external reference that was pinning it down But the thing about oh, I was standing on this corner and this person was with me and so on all that stuff drifts around Yeah, most people when they asked them a year later what they said was they saw a giant black man fly into the towers and bring it down Now we're talking about our visual perception and recognition But this synesthesia thing where you see a number but you visualize it as a color or you have a word and you taste it What is our?

a perception condition? How is that? Is this, okay, so it's something, am I correct me if I'm wrong, please? Is there something like 20% of the population could be? That's huge. Yeah, so is this now an evolutionary upgrade in how our mind performs? Okay, let me correct a couple things. It's not 20% of the population. No. But it, you know, so probably about 3% of the population has synesthesia. Not 22. Not 20. Yeah, no problem. There's certainly no reason to think that it's an evolutionary upgrade as such or that we're moving that way as a population. So let me explain. So what happens in every generation is that mother nature takes as much play in the wheel as she's able to. With every dimension that she can, she tries people shorter and taller and wider and this and that, and she tries everything out, including in the brain. So with synesthesia, there's essentially just a little bit more crosstalk between neighboring areas of the brain, sort of more porous borders between neighboring countries. And so as a result, people have a slightly different perception. But the reason, so synesthesia is a really important inroad. I've been studying this for 25 years in my lab. It's an important inroad into consciousness and how one tiny tweak, we've been studying the genetics of synesthesia as well. One tiny genetic tweak, Leads to a totally different way of experiencing reality, but it's not like synesthetic people are, let's say, more creative because if every time you look at the letter J, you have an experience of purple and the letter M triggers green and so on, that doesn't make you more creative. That just means you think, you know, J is purple and M is green and so on. So it may be that synesthetes have a very tiny advantage in some kind of memory tasks. So if I ask you to remember my phone number, maybe you'll forget it, but if you're synesthetic, you'll say, oh, I remember it had this nice autumn pattern

to it. And so then you're able to pull it up a little bit better. But it's, you know, especially in modern times, that's not too much of an advantage. So it's just an alternative perceptual reality. And I've been studying lots of different versions of this, just as one example. Have you guys heard of Afantasia and Hyperfantasia? Have you guys heard of this? No. So other than the Disney movie, no, the Fantasia. Okay. So this is spelled with a pH. So it turns out it turns out that If I ask you to picture something like an ant crawling on a red and white tablecloth towards a jar of purple jelly, some of you will see that clearly like a movie in your head. You can really I just did. OK, great. And others of us.

don't have really any visual at all. There's just no visual at all. And the population is spread over the spectrum pretty evenly. So if you can see, like Chuck apparently did, something super clearly, maybe euthanial, you know, like... Yes, complete. So that's called hyperfantasia, meaning your, you know, your imagery or internal imagery is really clear. I happen to be on the a-fantasia end of the spectrum where I don't really see anything. It turns out a friend of mine, Ed Katmall, who's the guy who started Pixar films, who has all the patents for ray tracing and all that stuff, he's a-fantasia. He got really interested in this topic some years ago and tested all the people at Pixar at his company. And it turns out that most of his best directors and animators are a-fantasia, meaning they don't picture anything on the inside. Now that seems like a big surprise, but Yes. What I've long hypothesized on this issue is that the A-fantasia kids become better artists. Why? Because the hyper-fantasia kid, the teacher says, draw a horse when you're in third grade. So you draw a horse because you think you know what a horse looks like and you probably do. But the A-fantasia kid says, God, I really have to study the model and look at my page and have a dialogue with the page, with the pencil.

And they get better and better over time at being artists because they really are having to concentrate on this. And so what you see when you look at these artists at Pixar is that they really are having a dialogue with the page and moving things around until it looks right. Well, you're saying, so you're saying a painter, a painter, not a house painter, but a fine arts painter, if they're a fantasic, they have to look If they have to look at what it is they're painting at all times because they can't picture it in their head. So that

makes sense because that person is going to be given to hyper detailed information. Whereas a person who can picture things in their head will do just that because you've got to think about it. The people at Pixar are working not with a paint and a brush. They're actually working with computers.

and numbers and that type of thing so and layering image on top of image on top of image that's how they build an image and they're doing it to match reality whereas somebody like Monet is probably just either near-sighted Or everything's out of focus. Everything's out of focus. Or they're just picturing this world that doesn't exist and then painting from that thing that they see in their head. OK. So, you know, I mean, that's just that's the that's the impression that I'm getting from that. I have no empirical evidence of this at all. I'm just I'm conjecturing right now. Yeah, I would say a couple. I mean, remember that Monet and all these guys, they sit in front of their landscape models and they're painting the Rouen Cathedral and so on from the from looking at it and doing it that way. The other thing to note is that, of course, Mone, it wasn't that he was near-sighted, it's that the camera had been invented and painters were trying to figure out how to distinguish themselves. Yeah, that's why you have impressionism. No, absolutely. That's why you have impressionism, because we had, reality had finally been replicated in a split instant. However, I will say this, as a person, it's very well known by the audience that I went away and did a week of ayahuasca and I had an experience where I did something that most of the people there didn't do, which was I hallucinated with my eyes open, where most of the people, they closed their eyes to see what they were seeing. We were having the same experience, but differently. The things that I was seeing was in the room with me that was, you know, it was really crazy, but, you know, I'm not sure if that has anything to do with what you're talking about, aphantasic or hyperphantasic or not.

Well generally what what's interesting about hallucinations which are always with the eyes open the interesting about hallucinations is that your brain is taking whatever little bits and drabs of data it has and putting together a story about what's going on out there and. I mentioned before that all of the stuff is, essentially all of it is internal activity is how your brain constructs your reality and you don't need your eyeballs at all to see

because you do this every night. That's what dreaming is. You're having full rich visual experience with your eyes closed because of the internal activity in your brain. Meaning that what you experience as vision when you're awake is really just awake dreaming. That's the way to think about vision.

And it turns out that your hallucination is just something, you know, just something in between there. You're a dreamer, Chuck. I'm a daydreamer. So, David, I have a philosophical question coming to this as a physicist where I care about what is objectively true, things that instruments that can determine about the natural world that bypass your senses and thereby separate what is objectively true from what is experienced to be true. But so my philosophical question is, if the letter M is green and the letter J is purple to a synesthetic person, who am I to say that my experience of color is the correct one and that their experience of color is crosswired? Who has the access to what is objectively true?

in the absence of a third way of verifying. If we're just humans out there in the Serengeti. Yeah, a couple layers to this, Neil. So first of all, what we experience is some sort of reflection of reality under normal circumstances, unless you're on ayahuasca or have dementia or 100 other things where you can pick up on the signals incorrectly and get them confused with memories and so on. But what has made our species successful is that we're able to agree. Hey, the coffee cup is over there, and the lion is over there, and so on. Now, it turns out, though, that, I mean, obviously, something like color is a construction of the brain. As you know, all we have is electromagnetic radiation of different wavelengths. And this, of course, leads to the old question that we all asked ourselves in college about, you know, how do I know that what I see is red is what you see is red and you and you? And the answer is it may well not be, and it doesn't matter. As long as we can transact and negotiate in the outside world, And I say, hey, can you please pass the red thing? Then we're fine. Our mothers teach us call that thing red, and so you call it red. And it doesn't matter if we're seeing the same thing on the inside. And there's no real reason to assume that it is the same for us. But aren't there members of the human species that actually can see more colors? What have they got? There's two and four. Tetra-chromatic. Thank you, tetra-chromatic.

So, most of us have what's called trichromatic vision because we have three types of color photoreceptors in our eyes, in our retina. But some women, because of a mutation on the X chromosome where they have two of them, they end up with four flavors of color photoreceptors and they see much more color than we do. It doesn't make it more real in a sense. Color is just a way of us tagging something so that we have an immediate perceptual experience. Instead of me saying, oh, I think that's 440 over there and that's 770 over there. It's just colors a way of having an immediate perceptual experience of the ripe fruit against the green leaves instead of tagging the wavelength in some other way that takes longer to figure out.

Well, if there's a subtle color difference between a poisonous berry and a healthy berry, it'd be nice to be able to see that color difference if I'm tetra. Tetra chromatic, yeah. Tetra chromatic. Yeah, yeah, it would be nice I don't think we have enough evidence to know that tetrachromatic women have have survived better They're very rare and it comes from mutation and if it were such a good mutation then maybe we would all have it Yeah, that makes sense David so with our perception of all different things ie our reality There's that conversation that people have about time where it seems to be stretched or it's short or whatever it is. And I know you've conducted a number of experiments, but where is our perception of time and how do we think it gets stretched and a moment in time is a lot longer than it actually was?

Okay, first in the answer to where, it's not in one location of the brain. It smeared out and in a minute it'll become clear why, which is, so I ended up collecting hundreds of reports from people who said that time seemed to go in slow motion when they were in a gunfight or a motorcycle accident or their child fell in the lake and they had to run after them and so on. There's all kinds of situations that are very dangerous to ourselves or something, someone we care about, where it seems like time runs in slow motion. And this had happened to me too when I was a kid and fell off of a roof. It seemed to take a long time. So I got really interested in this. And when I became a neuroscientist, I ran what is the first and only experiment that's ever been run on this. So I dropped volunteer subjects from a 150 foot tall tower Are you sure they were volunteers, David? They were. It took me eight months to get this approved, actually, by the committee to do the

experiment. So I dropped them. They're caught in a net below going 70 miles an hour. It's safe, but very, very scary to drop through the air like that. 70 miles an hour, 7-0. 7-0, exactly. Yeah. And the net has a big slew, so it slows you down. But it is so scary falling backwards from that height.

So what I did is I built my students and I we built this thing that goes on the wrist and flashes information at you at a certain rate and what we were testing is do you actually see in slow motion when you're in fear for your life and I'll skip many of the details, but the punchline is That you do not you don't actually see in slow motion what happens instead is when you're in fear for your life you are writing down tons of memories so that when you say what just happened, what just happened, which is all consciousness is, is probing on the recent memories, you've got all this memory to draw on. You say, oh my gosh, this is, so someone in a car accident says, oh yeah, let's see. I watched the hood crumple and I watched the rear view mirror fall off and I was looking at the expression on the other driver's face. So that must have taken four or five seconds because I've got all this memory that I can draw upon, but it's a trick of memory.

Wait, David, can I give an example? My first experience with this, which happened in a much safer environment than falling off a building. I think I've conveyed this story on a previous episode. When I was 14, maybe 15, I was driven across the United States nonstop from New York City to the Mojave Desert. And I'd never been west of Pennsylvania before. It was two consecutive days. We never stopped. Stop for gas. At the end of those two days, I felt like I had left New York a week earlier, not two days earlier. Because my memory had so many experiences in commensurate with any 48-hour period I had ever previously lived. And so my brain trying to make sense of it stretched it out, and I said, man, I must have left New York a week ago when it was only 48 hours ago. Is that a similar case here, David? Except my life was not at risk. That's exactly right. It turns out that most things, you're not catching it all.

We mentioned earlier that memory is the sieve. And so what happens in an emergency situation is that that's all memory is for. I mean, memory is there just so you can write down what happened when everything hit the fan so that you can avoid that situation again if you live. And so if you live, if you

don't do the experiment again. That's exactly right. So that's when you write everything down and your brain doesn't have a way. Look, Whenever you're judging duration, it's just a matter of how much memory you can pull up. So Neil on your road trip. Yeah. He just had lots of memory. Let me mention two things here. One is, this is why time seems to speed up as you grow older. It's because when you're a kid, everything is novel and you're writing down tons of memories. And when you get to the end of a day, there's just a ton of stuff that happened. When you get to the end of a summertime, You've gone to camp, you've gone hiking. There's all this new stuff. It's all new. You've written down tons of stuff. And when you're asked, what did you do? There's all these landmarks. But once you're an adult, your internal model of the world has developed pretty well. And there's not that much that's really novel. Now, sometimes I hear people say, oh, well, maybe it has to do with what fraction of your life something is. Now, a summertime is only a small fraction of our life. Whereas when you were a kid, It was a large fraction, but that's not it because you can go on a road trip or Neil, if you took, let's say a trip to the moon now, that is so novel and you would write everything down and you would come back on a Monday morning and say, I went to the moon this weekend and it would seem like from Friday was forever ago. Right. Got you. So David, go back to the high adrenaline events. Is it the adrenaline that triggers?

The amygdala to then dump this data rich messaging into the brain It's actually the signals from the amygdala that go tell the adrenaline to start dumping, which takes care of stuff with your muscles and so on. But yes, the amygdala, you can think of it as your emergency control center that says everybody pay attention to what is happening right here because normally your brain is doing 50 million things and you're thinking about what you're doing tomorrow and the vacation that's coming up and your grocery list and the call you need to make and whatever. But when the amygdala Kicks in the gear. It says everybody drop that and pay attention over here. And that's why you're writing stuff down with such density Wow, that is so cool. That is completely counterintuitive to what most people might think Yeah, but but it makes complete sense. Yeah, it makes great sense I'm so glad you brought that up. That's a whole new way of looking at time You know, even though we know that time is a perception to begin with unless you know your astrophysicist in which time is an actual thing that can actually slow

down because of gravity and your proximity to the speed of light. It slows down as viewed by others. As viewed by others. Correct. Your own internal state is the same.

Right. Well, this is what I was going to say is in physics, there's this notion of relativity, but there's a there's a neural relativity with time also, but it's a very different sort of thing. But it's this issue of if I'm in a situation and I'm really scared and you're looking at the same situation, but you're not scared, we're going to have a totally different sense of how much time had passed. So wait, say that again. The neural relativity is if we're both in the same We're both in one situation and you're in another situation. I'm scared in mind, you're completely comfortable. Is that what you're saying? What I'm saying is we're in the same situation, but for whatever reason, you've seen this thing before. Oh, okay. So we're jumping out of a plane and I'm strapped to you. You've done it a hundred times. It's my first time. Exactly right. That's exactly right. Chuck is screaming the whole way. Absolutely. My experience of time is...

That's my experience. David, we discussed a couple of your experiments, but there is one that I came across that actually doesn't involve being pushed off a high tower, jumping out of a plane. And I think our audience would like to be able to participate in one of your experiments. So if I've got this right, it's the looking in the mirror at your left eye, then your right eye. Can you break that down and explain? Because our audience will be able to do this themselves. Do the following, stand in front of a mirror and stand right up close to it, right up against the mirror and with both your eyes, you look at your left eye, then you look at your right eye, then you look at your left eye.

Okay, now watch somebody else do this also. Watch them look at their left eye and right eye and so on. When you're watching someone else, you can see the eyes move. The eyes make these big jumps. It's called saccade, and that's when they shift from looking from one eye to the other. Okay, easy. But when you look at your own eye, you never experience that shift, that jump that your eyes make. Instead, you feel like you're looking at your left eye, then instantly it switches and you're looking at your right eye. then instantly it switches and you're looking at your left eye. And the question is,

what the heck is going on while your eyes are making this big shift and jumping from one place to the other? How come you don't experience that? Well, we have these gaps in time all the time where you just never experience that. And that's because all you ever see is your internal model of the world. And there's all this weird Post processing that goes on like like an editing software program. So it just fills in that gap. So you think you're looking your left eye and then you're immediately seeing your right eye. But what that exposes that simple experiment is the degree to which time isn't experienced exactly as things are happening outside world. I didn't want to learn that.

I don't want any knowledge that my experience is not connected to an objective reality. So could you take that back pretend you never said that? I wish I could. Well, I mean, for 25 years, I've been studying illusions of time. There are actually lots and lots of different things about time, you know, not only gaps in time and all the things going faster and slower, but also something happening before something else. Here I'll just tell you one experiment we do I can tell this real fast Which is we have you hit a button and that causes a flash of light so every time you hit the button There's a flash of light now what we do is we inject a little delay Let's say 20 milliseconds So you hit the button and the flashlight happens 20 milliseconds later and we make this delay bigger and bigger each time and if we do that slowly Then you know you can be hitting the button There's a 200 millisecond delay before the light flashes and your brain recalibrates so that it doesn't see the delay at all. It thinks that it's instant. Now I remove the delay. So after you've gotten used to this 200 millisecond delay, I remove it. So now you hit the button and the light flashes and you say, whoa, I didn't do that. It flashed before I hit the button. What? Yeah. Wow. There's a reversal of action and effect because your brain is recalibrating. Yeah, recalibrating. That's insane. So your brain is a liar. It's just a liar all the time. Your brain is just It's trying to put together the best story you can. And if there's a delay in some system, then it says, oh, that's cool. I'll just adjust that. I know I'm the one hitting the button. So I'm going to assume there's this delay for some other reason. So the first time I saw this, this illusory reversal of action and effect.

I thought, you know what? That's exactly what happens in schizophrenia. Because my participant said, well, I didn't do that. That wasn't me. And what happens in schizophrenia is what's called credit misattribution, where somebody will do something and say, that wasn't me. I wasn't the one who did it. And so that led me down this road that I've been pursuing for years, which is that schizophrenia is actually fundamentally a disorder of time perception. And that if you're not getting your timing signals lined up exactly right, then schizophrenia is the obvious thing that happens. For example, You're always talking to yourself. You're always generating an internal voice and listening to it But if you get the timing of that just slightly wrong so that you think you heard the voice before you generated it That's an auditory hallucination. You have to attribute voice to somebody else And someone's talking to you. Yeah, yeah, okay. That's a that's a brilliant take man Yeah, that's a really recent is there any research going on to kind of sort of re-engineered or sort of back-engineered a solution to these sort of issues for people with mental health problems? Yeah. So that's exactly what I'm working on. That's exactly what I'm trying to figure out. I mean, here's the dream. It'll take me some years before hopefully I get there. But the dream is that you take a person suffering with schizophrenia and you have them play a video game that recalibrates their timing and then their auditory hallucinations go away. So we'll see if I mean, I can't say that works, but I hope so. I hope in five years we can get back together and celebrate that. David.

If I'm ever in Palo Alto, I want to visit your lab and I want you to do that light delay experiment on me, because I refuse to believe that I would function that way. I'm in denial, okay? And I would need the experiment. I refuse to believe it. It's a super simple experiment. By the way, I presented this at a university and afterwards a guy came up to me and said, you know, this explains something, which is we just got a new telephone system installed. It turns out, by the way, this is something that I had studied also. It turns out when you hit a key on the keyboard, there's like a hundred millisecond delay before the letter shows up on your screen. So anyway, he said, look, I'd be typing in my computer and then I go to my phone and type in a number.

It turns out on a telephone, there's essentially no delay and especially on these new telephones that he just got installed. So what happened is when he dialed the last number, he had the impression that the phone started ringing before he hit the last number because his brain got used to the delay on the keyboard. So now when he went to the telephone, which had no delay, he had this illusion of the reversal of action in effect. I'm going to naively declare that I'm not susceptible to that. That's how much I believe my own objective reality. And so I have to prove it. I'm going to find you. I'm so glad that we have this recorded. We'll have to hold it out when I show you this. I'm going to find you. Neil's sound like somebody on the internet telling him about vaccines right now. Right. You've got a good brain, Neil, but it's like everybody else's.

So David, earlier you mentioned certain evidence of consciousness, some functions of the brain. There's no end of books on a shelf written by people in your world who declare that they have explained consciousness. And as an academic, evidence that people don't know anything about a subject is brought forth by the fact that people keep writing books on that subject. That's the evidence that is not fully explained. Because if it were explained, people would stop writing books because it would be explained. So it's where do you feel about this? And is consciousness even something that needs an explanation? Or is it just some emergent property of having neural activity? It definitely needs explanation because consciousness is, you know, our subjective impression of having this world and the pain and the beauty and the agony and the ecstasy, all of that internally that I am a self and all that, that is my conscious experience. But somehow when we look at the brain and the body, it's just a bunch of cells. It's cells doing very cool neural networks, the original neural networks and there's chemicals and there's all kinds of stuff going on. But fundamentally it's a wet biological machine. And when we look at other sophisticated machines like the laptop on my table here, that's also made up of lots of pieces and parts, but we don't suspect that it's having internal experience. And so it is a real question, and I would say it's the deepest mystery that we have in neuroscience is why does it feel like something to have all these pieces and parts going? And by the way, it's very easy for conscious experience to change. So when people put alcohol in their system, or drugs of various sorts, or...

All that changes their conscious experience. If you hit your head and go into a coma, your conscious experience just went away, even though the rest of your body is fine and working, your consciousness has gone away. Anyway, somehow we think it's an emergent property of the algorithms that are being run here, but it wouldn't be sufficient to just end there because we don't know why that happens. Part of the reason that this is so important to answer this question. We have a new reason why this is important is because now we're dealing with this new world of AI and we're asking the question, how would we know if an LLM were conscious or the next, you know, a large world model becomes conscious? Because then you don't want to turn off the computer. Does that count as murder? All kinds of new questions that crop up with this. And that's why we're trying to understand why that happens.

By the way, the in the film, I didn't read the original story. Forgive me. But so in the movie, I robot original story by Isaac. Yeah, there are these decommissioned robots and they're put in this truck and you go inside the truck and the robots tended to mingle with each other rather than just stay where you put them. They've been decommissioned. So there's no. There's nothing actively programmed going on in them. And that was explained by the fact that the software, since it was always just simply updated and not reprogrammed from scratch, there were these lingering software pathways within them that were long gone out of the needs of later functioning, but were still there operating.

And when I heard that and I learned that, all I could think of was evolutionarily, we have so much baggage of what it is to be alive, just leftover from millions of years that maybe this leftover wiring that no one wants in our daily existence still operates in our own sense of self without us even trying to make that happen. Yeah, that's certainly a possibility. One issue, though, is that consciousness does seem to be useful because it's what allows you to have a high-level operating system and say, look, I'm made of 30 trillion trillion cells, but I need to know whether I'm going left or right around that tree. And so I'm going to make a decision that takes care of all these systems and all this stuff and coordinates as one being. And by laying down memory and saying, oh, that memory happened to me as a individual. And that's therefore why I can predict that I should do this in the future and

so on. You get a lot more use out of the whole system this way. And so consciousness appears to be very useful.

It has allowed us as a species to coordinate and cooperate to build whole civilizations. So, you know, it's more than an evolutionary piece of baggage that we'd want to get rid of like the appendix. It's something that's really useful. But call me when you figured it out and we could ignore all subsequent books written on that subject. I'll let you know. Yeah, I think that's all the time we have. Damn. Fascinating stuff. Great. Well, thanks guys. Great to talk with y'all. All right, David. Delight to have you on StarTalk. And stop wasting time talking to us. Get back to the lab and figure out our inner cosmos. Go scare the crap out of more people. Right away. All right, take care y'all. Great to see you. This has been another installment of StarTalk Special Edition.

That was all about the inner cosmos. Chuck, always good to have you, man. Always a pleasure. And Gary. Great show. Loved it. All right. This is Neil deGrasse Tyson, your personal astrophysicist. As always, keep looking up.